

Problem Set 1 – Diagnostic

Session 1 · Course overview and school math refresher

HOW TO ANSWER Every answer carries one sentence of justification. A value on its own is not a solution – say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

Part A · Algebraic care

1 EXPONENTS

Simplify $\frac{27^{2/3} \cdot 3^{-2}}{9^{1/2}}$, stating which exponent law you use at each step.

SOLUTION $27^{2/3} = (3^3)^{2/3} = 3^2 = 9$ by $(a^m)^n = a^{mn}$; $9^{1/2} = 3$. So the expression is $\frac{9 \cdot 3^{-2}}{3} = \frac{1}{3}$.

2 RADICALS

Decide whether $\sqrt{x^2} = x$ holds for all real x . If not, give the correct identity and a value of x that breaks the false one.

SOLUTION False. $\sqrt{\cdot}$ denotes the non-negative root, so $\sqrt{x^2} = |x|$. At $x = -3$: $\sqrt{9} = 3 \neq -3$.

3 IDENTITIES

Prove that $(a + b)^2 - (a - b)^2 = 4ab$ for all real a, b . State explicitly why your argument covers all real a, b and not just the ones you tried.

SOLUTION Expand both squares: $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$. Every step is a ring identity valid for arbitrary reals, so no case analysis is needed – the letters were never assumed to be particular numbers.

4 COMPLETING THE SQUARE

Write $2x^2 - 12x + 23$ in the form $a(x - h)^2 + k$ and read off its minimum value and where it occurs.

SOLUTION $2(x^2 - 6x) + 23 = 2((x - 3)^2 - 9) + 23 = 2(x - 3)^2 + 5$. Since $(x - 3)^2 \geq 0$ with equality only at $x = 3$, the minimum is 5 at $x = 3$.

Part B · Absolute value and inequalities

5 ABSOLUTE VALUE

Solve $|3x + 1| \geq 4$. Mark each step $\iff$ or $\implies$ and say why.

SOLUTION $|u| \geq 4 \iff u \leq -4$ or $u \geq 4$ (definition of $|\cdot|$, a case split, not a loss of information). With $u = 3x + 1$: $3x \leq -5$ or $3x \geq 3$, i.e. $x \leq -\frac{5}{3}$ or $x \geq 1$. Dividing by $3 > 0$ preserves the direction, so every step is an equivalence.

6 RATIONAL INEQUALITY

Find all real x with $\frac{2}{x-1} \leq 1$. Handle the sign of $x - 1$ explicitly; do not multiply through by an expression of unknown sign.

SOLUTION Case $x > 1$: multiplying by $x - 1 > 0$ gives $2 \leq x - 1$, so $x \geq 3$. Case $x < 1$: multiplying by $x - 1 < 0$ flips the inequality, $2 \geq x - 1$, i.e. $x \leq 3$, which holds throughout $x < 1$. At $x = 1$ the expression is undefined. Answer: $x < 1$ or $x \geq 3$.

7 AM-GM

Show $x^2 + y^2 \geq 2xy$ for all real x, y , and determine exactly when equality holds.

SOLUTION $x^2 + y^2 - 2xy = (x - y)^2 \geq 0$ because squares of reals are non-negative. Equality holds iff $(x - y)^2 = 0$, i.e. $x = y$.

8 FINDING THE ERROR

A student writes: "Since $\frac{1}{x} < 1$, multiply both sides by x to get $1 < x$." Identify the false step, and give a value of x for which the premise is true but the conclusion is false.

SOLUTION Multiplying by x is illegal because the sign of x is unknown; if $x < 0$ the inequality must flip. Take $x = -2$: $\frac{1}{-2} = -0.5 < 1$ is true, but $1 < -2$ is false.

Part C · Write like a mathematician

9 PRECISION

Each sentence below is sloppy. Rewrite it so it is unambiguous and true.

- a. "The square root of a number is positive."
- b. "If you square both sides of an inequality you get an equivalent inequality."
- c. " $x^2 = 4$, so $x = 2$."

SOLUTION (a) For $a \geq 0$, $\sqrt{a}$ denotes the unique non-negative real whose square is a .
(b) If both sides are non-negative, squaring is an equivalence; otherwise it is only an implication. (c) $x^2 = 4 \iff x = 2$ or $x = -2$.

10 REFLECTION

In no more than eight sentences: pick one rule you were taught at school without a reason (for example the quadratic formula, or "you can't divide by zero"). State the rule precisely, then say what would have to be true for it to fail. You are not being asked to prove it.

WHAT A GOOD ANSWER DOES States the rule precisely, then gives a genuine failure condition rather than a restatement of the rule, and says it clearly. There is no single correct answer.